

tion—approximately 13 percentage points apart in on-the-job computer use.

We have estimated many variations of this basic model to verify its robustness, including specifying the dependent variable as the level or percentile rank of industry computer use in 1984, 1997, or the average of both; scaling the routine task share measure in percentiles of the 1960 task distribution; calculating the routine task share index using task percentiles rather than task levels; and replacing the routine task index with its logarithm. These many tests provide robust support for the first proposition of our theoretical model: demand for computer capital is greatest in industries that were historically routine task intensive.

III. TRENDS IN JOB TASK INPUT, 1960–1998

Our model implies that the rapidly declining price of computer capital should have reduced aggregate demand for labor input of routine tasks and increased demand for labor input of nonroutine cognitive tasks. This section analyzes the evidence for such shifts.

III.A. Aggregate Trends

Figure I illustrates the extent to which changes in the occupational distribution over the period 1960 to 1998 resulted in changes in the tasks performed by the U. S. labor force. This figure is constructed by pairing the selected DOT 1977 task measures with Census and CPS employment data for each decade. By construction, each task variable has a mean of 50 centiles in 1960. Subsequent points depict the employment-weighted mean of each assigned percentile over each decade.¹⁴

As is evident in the figure, the share of the labor force employed in occupations that made intensive use of nonroutine analytic and nonroutine interactive tasks increased substantially during the last four decades. Although both of these measures of nonroutine tasks increased in the 1960s—that is, during the

14. We do not impose an adding-up constraint across task measures—whereby total task allocation must sum to one within jobs or time periods—since this structure is not intrinsic to the DOT. It is therefore possible for the economywide average of total task input to either rise or fall. This over-time variation is modest in practice. The mean of all five task measures, equal to 50 by construction in 1960, rose slightly to 52.5 in 1980, and fell to 51.2 in 1998.

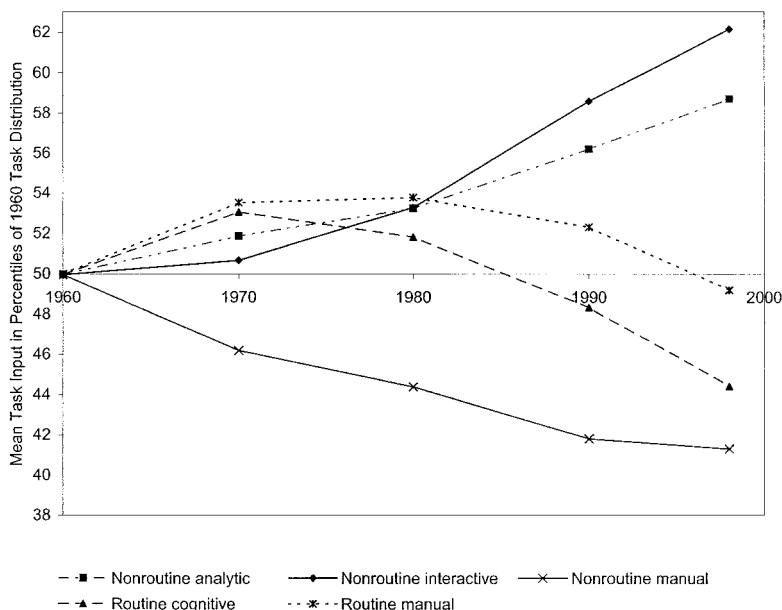


FIGURE I

Trends in Routine and Nonroutine Task Input, 1960 to 1998

Figure I is constructed using *Dictionary of Occupational Titles* [1977] task measures by gender and occupation paired to employment data for 1960 and 1970 Census and 1980, 1990, and 1998 Current Population Survey (CPS) samples. Data are aggregated to 1120 industry-gender-education cells by year, and each cell is assigned a value corresponding to its rank in the 1960 distribution of task input (calculated across the 1120, 1960 task cells). Plotted values depict the employment-weighted mean of each assigned percentile in the indicated year. See Table I and Appendix 1 for definitions and examples of task variables.

precomputer era—the upward trend in each accelerated thereafter. By 1998, nonroutine analytic task input averaged 6.8 centiles above its 1970 level and nonroutine interactive input averaged 11.5 centiles above its 1970 level.

By contrast, the share of the labor force employed in occupations intensive in routine cognitive and routine manual tasks declined substantially. Between 1970 and 1998, routine cognitive tasks declined 8.7 centiles and routine manual tasks declined by 4.3 centiles. Notably, these declines reversed an upward trend in both forms of routine task input during the 1960s. For routine cognitive tasks, this trend reversed in the 1970s, and for routine manual tasks, the trend halted in the 1970s and reversed in the 1980s.

Finally, the share of the labor force employed in occupations intensive in nonroutine manual tasks showed a secular decline across all decades. This decline was most rapid in the 1960s, and slowed considerably in subsequent decades.

Panel A of Table II provides the corresponding means of task input by decade, both in aggregate and by gender. For both males and females, there are pronounced shifts against routine cognitive, routine manual, and nonroutine manual task inputs, and pronounced shifts favoring nonroutine analytic and interactive inputs. These shifts, however, are numerically larger for females. Given the rapid entry of women into the labor force in recent decades, it appears plausible that demand shifts for workplace tasks would impact the stock of job tasks more rapidly for females than males [Goldin 1990; Weinberg 2000; Blau, Ferber, and Winkler 2002, Chapter 4]. To assess the importance of these gender differences, we estimated all of our main results separately for males and females. Because we found quite similar results for both genders, we focus below on the pooled gender samples.

To complete the picture provided by the decadal means, Figure II depicts smoothed changes in the density of the two routine and two nonroutine cognitive task measures between 1960 and subsequent decades. Three series are plotted for each task measure. Two depict extensive margin task shifts at approximately twenty-year intervals. These are measured using the 1977 DOT task measures paired to the 1960, 1980, and 1998 employment data. The third series adds intensive margin task shifts by pairing the 1991 DOT task measures with the 1998 employment data. By construction, task input is uniformly distributed across all percentiles in 1960. Hence, the height of each line in the figure represents the difference in the share of overall employment in 1980 or 1998 at each centile of 1960 task input.¹⁵ To conserve space, we do not provide a plot of the nonroutine manual measure, since it is not the subject of subsequent analysis.

As shown in panels A and B of the figure, the distribution of nonroutine analytic and nonroutine interactive task input shifted markedly rightward after 1960. In particular, there was substantial growth in the share of employment requiring nonroutine task input above the 1960 median and a corresponding decline below

15. We apply an Epanechnikov kernel with bandwidth $h = 0.90\sigma n^{-1/5}$, where n is the number of observations and σ is the standard deviation. For our samples, this yields bandwidths between 5 and 7 centiles.

TABLE II
MEANS OF TASK INPUT BY DECADE AND DECOMPOSITION INTO WITHIN AND BETWEEN INDUSTRY COMPONENTS, 1960–1998

A. Weighted means of economywide task input by decade (percentiles of 1960 task distribution)															
	1. Nonroutine analytic			2. Nonroutine interactive			3. Routine cognitive			4. Routine manual			5. Nonroutine manual		
	All	Male	Fem	All	Male	Fem	All	Male	Fem	All	Male	Fem	All	Male	Fem
1960 (Census)	50.0	54.6	37.0	50.0	58.4	26.4	50.0	48.6	53.8	50.0	43.2	69.2	50.0	56.5	31.6
1970 (Census)	52.5	57.5	41.5	51.1	61.7	27.5	52.2	49.8	57.5	54.0	44.9	74.4	46.9	54.1	30.8
1970 (Census)	51.9	56.4	41.8	50.7	60.8	28.1	53.1	50.8	58.2	53.5	44.7	73.3	46.2	53.2	30.7
1980 (Census)	54.9	58.2	49.2	55.4	63.3	41.8	52.9	50.0	58.0	55.1	44.6	73.3	44.0	53.2	28.1
1980 (CPS)	53.2	56.6	47.9	53.3	61.4	40.4	51.8	50.0	54.8	53.8	43.3	70.4	44.4	55.0	27.5
1990 (CPS)	56.2	57.4	54.6	58.6	62.7	53.0	48.3	48.1	48.6	52.3	42.8	65.5	41.8	53.1	26.3
1998 (CPS)	58.7	59.3	58.0	62.2	63.9	59.9	44.4	46.6	41.6	49.2	41.5	59.3	41.3	52.6	26.5
B. Decomposition of task shifts into between and within industry components for combined genders (10 × annual changes in mean task percentile)															
	Total	Btwn	Wthn	Total	Btwn	Wthn	Total	Btwn	Wthn	Total	Btwn	Wthn	Total	Btwn	Wthn
1960–1970	2.57	1.74	0.83	1.15	−0.34	1.49	2.20	1.14	1.06	4.01	2.39	1.62	−3.03	−2.28	−0.74
1970–1980	3.02	1.54	1.48	4.68	0.26	4.42	−0.14	0.33	−0.47	1.63	0.79	0.84	−2.25	−1.00	−1.25
1980–1990	2.97	0.92	2.05	5.31	0.52	4.79	−3.48	−1.42	−2.07	−1.47	−0.16	−1.31	−2.58	−1.27	−1.31
1990–1998	3.12	0.67	2.45	4.48	0.54	3.94	−4.88	−1.31	−3.57	−3.88	−0.38	−3.50	−0.63	−0.31	−0.31

Sources: *Dictionary of Occupational Titles* [1977], and all employed workers ages 18–64. Census IPUMS 1960, 1970, 1980, CPS MORG 1980, 1990, and 1998. Samples used for decadal changes in panel B are 1960–1970, 1960 and 1970 Census; 1970–1980, 1970 and 1980 Census; 1980–1990, 1980 and 1990 CPS MORG; 1990–1998, 1990 and 1998 CPS MORG. Two Census 1970 samples are used in panels A and B, one coded for consistency with the 1960 Census occupation codes and a second coded for consistency with the 1980 Census occupation codes. Data are aggregated to 1120 industry-gender-education cells by year and each cell is assigned a value corresponding to its rank in the 1960 distribution of task input (calculated across the 1120, 1960 task cells). Panel A contains the employment-weighted mean of each assigned percentile in the indicated year. Panel B presents a decomposition of the aggregate change in task input over the indicated years into within and between industry components for 140 consistent Census Industry Code (CIC) industries in manufacturing, 81 in nonmanufacturing. (See Table I and Appendix 1 for definitions and examples of task variables.)